

Difference-in-Differences in the Presence of Unknown Interference

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Motivation

- Difference-in-Differences methods are very popular in observational settings, when treatment assignment is confounded with unobservable characteristics.
- They rely (explicitly) on Parallel Trends (PT) type of assumptions and (implicitly) SUTVA (consistency + no interference).
- While the former has been largely studied and criticized, the latter has not received much attention.
- “There has been little attention paid to models allowing for such interference in the recent causal panel data literature to date.”
Arkhangelsky and Imbens (2024)

What is interference?

- A unit's outcome depends on the treatment status of other units with whom she interacts
- Interference/spillover effects may arise
 - general equilibrium effects
 - units interacting within and between groups or over a network
 - spatial proximity

Recent DiD literature does not even mention SUTVA

Difference-in-Differences with Multiple Time Periods*

Brantly Callaway[†]

Pedro H. C. Sant'Anna[‡]

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Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects[†]

By CLÉMENT DE CHAISEMARTIN AND XAVIER D'HAUTFEUILLE*

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Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects*

Liyang Sun[†] and Sarah Abraham[‡]

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Difference-in-differences with variation in treatment timing*

Andrew Goodman-Bacon*

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Motivation



Post



Kirill Borusyak

@borusyak



It's common to hear "regression only identifies relative effects". To me, that's not a good way to think about it. What do "relative effects" even mean?

Regression identifies **causal** effects but only under SUTVA. Without it, we model interference; GE models is one approach.

[Traducir post](#)



International and Monetary Economics Net...  @int_mon_e... · 11 sept.

Highly recommended!

"The 'missing intercept' problem with going from micro to macro" by Benjamin Moll and Oliver Hanney.

...

[Mostrar más](#)



Motivation

- “There are likely substantial [...] spillover effects on one’s mental health due to one’s peers’ social media use. We employ the term “general equilibrium effect” to indicate that *our estimates capture such indirect effects*, as well as any *direct effects*.” Braghieri et al. (2022)
- “For each group, we select the set of potential control municipalities to be the never-treated municipalities in one of the ten regions that experienced a CCD other than r . We require the control group to be in a region other than r to *avoid contamination from spillover effects*.” Fenizia and Saggio (2024)
- “For example, scholars at universities that were treated later may be inspired by earlier universities switching to coeducation and began producing gender-related research prior to coeducation. *This would bias our estimates upward*.” Truffa and Wong (2025)
- “To quantify the aggregate and welfare effects of the policy, we specify and estimate an industry equilibrium model of conglomerate production. The model accounts for conglomerate and market spillovers, matches the estimated impacts of the policy, and clarifies the *interpretation of the difference-in-differences estimates in the presence of both types of spillovers*.” □ Chen et al. (2025)

DiD settings and interference

- What are DiD estimates capturing in the presence of interference?
- (When) are DiD estimates lower bounds for ATT effects?
- Our contribution
 - Clarify what DID estimands are and restate the PT assumption with interference, being agnostic about the pattern of interference
 - Under what assumptions 1) on the trend in the absence of any intervention and 2) on the sign of the total (or "direct") and the spillover effects can DID estimands be interpreted as lower (or upper!) bounds of total effect
 - Revisit Card and Krueger (1994) minimum wage example

2 groups 2 periods DiD

- A sample of size N , each unit indexed by $i \in \{1, \dots, N\}$.
- All units observed in two periods $t \in \{0, 1\}$.
- W_{it} binary treatment indicator of unit i at time t .
- At time $t = 0$, no unit is treated ($W_{i0} = 0 \quad \forall i$).
At time $t = 1$, some units are treated ($W_{i1} \in \{0, 1\}$).
- G_i is the group indicator, 1 if unit is treated at $t = 1$, 0 otherwise.

Assignment Matrix

- Let $\mathbf{W}$ denote the assignment matrix:

$$\mathbf{W} = \begin{pmatrix} W_{10} & W_{11} \\ W_{20} & W_{21} \\ \vdots & \vdots \\ W_{N0} & W_{N1} \end{pmatrix} = (\mathbf{w}_0 \quad \mathbf{w}_1)$$

- Y denotes the outcome of interest.
- Potential outcomes are:

$$Y_{it}(\mathbf{W})$$

Decomposing the Assignment Matrix

- **Assumption 1:** No Anticipation

$$Y_{i0}(\mathbf{W}_0, \mathbf{W}_1) = Y_{i0}(\mathbf{W}_0, \mathbf{W}'_1) = Y_{i0}(\mathbf{W}_0) \quad \forall \mathbf{W}_1, \mathbf{W}'_1$$

- Because no one is treated at $t = 0$, we can simply write $Y_{i0}(\mathbf{W}_0) = Y_{i0}(\mathbf{0})$ and:

$$Y_{i1}(\mathbf{W}_0, \mathbf{W}_1) = Y_{i1}(\mathbf{0}, \mathbf{W}_1) = Y_{i1}(\mathbf{W}_1)$$

- We can decompose further the vector of treatment assignment at time t in the unit's own treatment, their group treatment, and the other group treatment:

$$Y_{it}(\mathbf{W}_t) = Y_{it}(W_{it}, \mathbf{W}_{G_{it}}, \mathbf{W}_{1-G_{it}})$$

Potential Outcomes

$$Y_{it}(\mathbf{W}_t) = Y_{it}(W_{it}, \mathbf{W}_{G_i t}, \mathbf{W}_{1-G_i t})$$

- **Assumption 2: Consistency**

$$Y_{i0} = Y_{i0}(0, \mathbf{0}, \mathbf{0})$$

$$Y_{i1} = W_{i1} \times Y_{i1}(1, \mathbf{1}, \mathbf{0}) + (1 - W_{i1}) \times Y_{i1}(0, \mathbf{0}, \mathbf{1})$$

Canonical DiD

Assumption 3: No Interference

$$Y_{it}(W_{it}, \mathbf{W}_{G_i t}, \mathbf{W}_{1-G_i t}) = Y_{it}(W_{it}, \mathbf{W}'_{G_i t}, \mathbf{W}'_{1-G_i t}) = Y_{it}(W_{it})$$

$$\forall \mathbf{W}_{G_i t}, \mathbf{W}_{1-G_i t}, \mathbf{W}'_{G_i t}, \mathbf{W}'_{1-G_i t}$$

Assumption 4: Parallel Trends

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0]$$

Estimand 1: Difference-in-Differences

$$DiD = \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 1] - \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 0].$$

Estimand 2: Average Treatment effect on the Treated

$$ATT = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid G_i = 1]$$

Identification Result

From Parallel Trends

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0],$$

we get the missing potential outcome for the treated units as :

$$\begin{aligned}\mathbb{E}[Y_{i1}(0) \mid G_i = 1] &= \mathbb{E}[Y_{i0}(0) \mid G_i = 1] + \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0] \\ &= \mathbb{E}[Y_{i0} \mid G_i = 1] + \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 0]\end{aligned}$$

So we can write the ATT

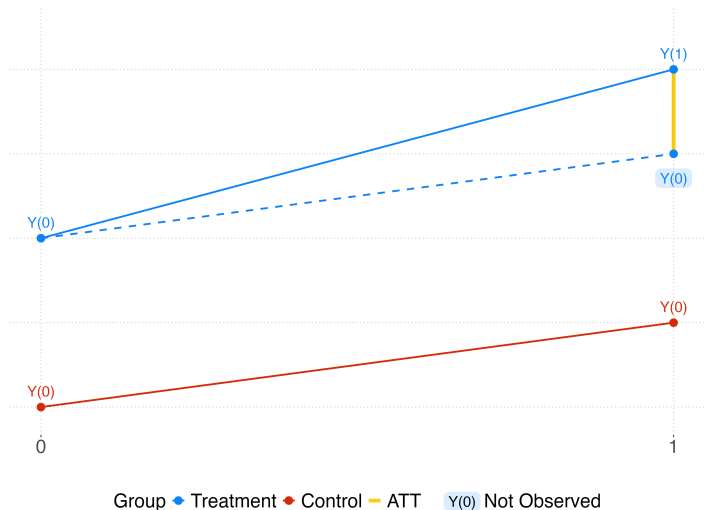
$$\mathbb{E}[Y_{i1}(1) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0) \mid G_i = 1]$$

as

$$\mathbb{E}[Y_{i1}(1) \mid G_i = 1] - \mathbb{E}[Y_{i0} \mid G_i = 1] - \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 0] =$$

$$\boxed{\mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 1] - \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 0]} \quad (1)$$

Difference in Differences



DiD with interference

Assumption 5: No interference between groups (partial interference)

$$Y_{it}(W_{it}, \mathbf{W}_{G_it}, \mathbf{W}_{1-G_it}) = Y_{it}(W_{it}, \mathbf{W}_{G_it}, \mathbf{W}'_{1-G_it}) = Y_{it}(W_{it}, \mathbf{W}_{G_it}) \\ \forall \mathbf{W}_{1-G_it}, \mathbf{W}'_{1-G_it}$$

Assumption 6: Modified Parallel Trends

$$\mathbb{E}[Y_{i1}(0,0) - Y_{i0}(0,0) \mid G_i = 1] = \mathbb{E}[Y_{i1}(0,0) - Y_{i0}(0,0) \mid G_i = 0]$$

The DiD estimand identifies a "total" average effect on the treated:

$$\mathbb{E}[Y_{i1}(1,1) - Y_{i1}(0,0) \mid G_i = 1] \quad (2)$$

Estimands under unknown interference

Estimand 3: Average total (or "direct") effect on the treated

$$\tau^1 = \mathbb{E}[Y_{i1}(1, 1, 0) - Y_{i1}(0, 0, 0) \mid G_i = 1]$$

Estimand 4: Average indirect (or spillover) effect on the control

$$\tau^0 = \mathbb{E}[Y_{i1}(0, 0, 1) - Y_{i1}(0, 0, 0) \mid G_i = 0]$$

Assumption 7: Parallel Trends under unknown interference

$$\begin{aligned} & \mathbb{E}[Y_{i1}(0, 0, 0) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0, 0, 0) \mid G_i = 0] = \\ & \mathbb{E}[Y_{i0}(0, 0, 0) \mid G_i = 1] - \mathbb{E}[Y_{i0}(0, 0, 0) \mid G_i = 0] \end{aligned}$$

What does the DiD estimand identify? I

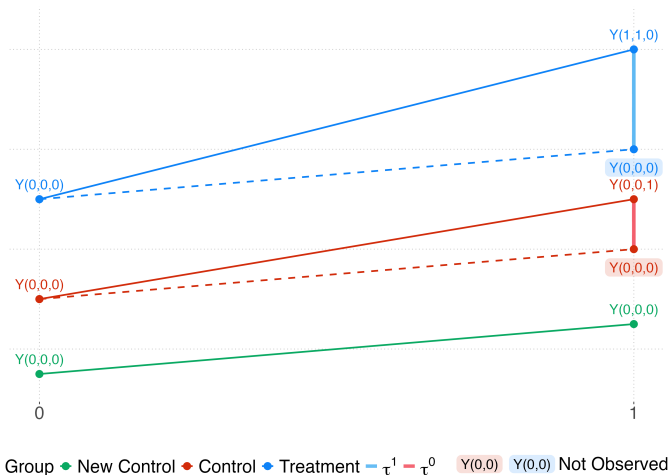
$$\begin{aligned} & \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 1] - \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 0] \\ &= \left(\mathbb{E}[Y_{i1} \mid G_i = 1] - \mathbb{E}[Y_{i1} \mid G_i = 0] \right) - \left(\mathbb{E}[Y_{i0} \mid G_i = 1] - \mathbb{E}[Y_{i0} \mid G_i = 0] \right) \\ &= \left(\mathbb{E}[Y_{i1}(1, 1, 0) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0, 0, 1) \mid G_i = 0] \right) - \\ & \quad \left(\mathbb{E}[Y_{i0}(0, 0, 0) \mid G_i = 1] - \mathbb{E}[Y_{i0}(0, 0, 0) \mid G_i = 0] \right) \\ &= \left(\mathbb{E}[Y_{i1}(1, 1, 0) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0, 0, 1) \mid G_i = 0] \right) - \\ & \quad \left(\mathbb{E}[Y_{i1}(0, 0, 0) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0, 0, 0) \mid G_i = 0] \right) \end{aligned}$$

What does the DiD estimand identify? II

$$\begin{aligned} &= \left(\mathbb{E}[Y_{i1}(1, 1, 0) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0, 0, 1) \mid G_i = 0] \right) - \\ &\quad \left(\mathbb{E}[Y_{i1}(0, 0, 0) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0, 0, 0) \mid G_i = 0] \right) \\ &= \underbrace{\mathbb{E}[Y_{i1}(1, 1, 0) - Y_{i1}(0, 0, 0) \mid G_i = 1]}_{\tau^1} - \underbrace{\left(\mathbb{E}[Y_{i1}(0, 0, 1) - Y_{i1}(0, 0, 0) \mid G_i = 0] \right)}_{\tau^0} \\ &= \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 1] - \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 0] = \tau^1 - \tau^0 \end{aligned}$$

Some way-out found in the literature

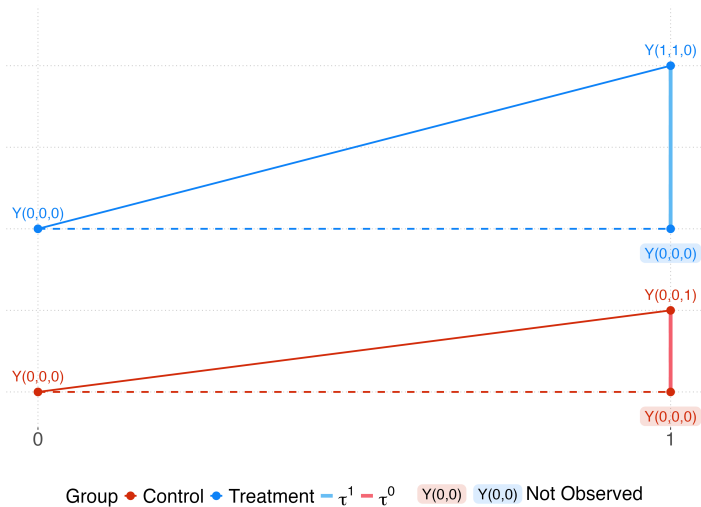
- Find a new control group, for which you observe $Y_1(0,0,0)$ (Hettinger et al., 2024; Butts, 2023).



Alternative Assumptions

- 1 Make assumptions on $Y_{i1}(0,0,0)$ - may lead to only partial identification, similar to Rambachan and Roth (2023).
- 2 Make assumptions on τ^1 and τ^0 .

Assumptions on $Y(0,0,0)$



Assumptions on τ^1 and τ^0

- Sign of the average indirect/spillover effect on the control

$$\begin{aligned}\tau^0 \geq 0 &\implies DiD \text{ is a lower bound of } \tau^1 \\ \tau^0 \leq 0 &\implies DiD \text{ is an upper bound of } \tau^1\end{aligned}$$

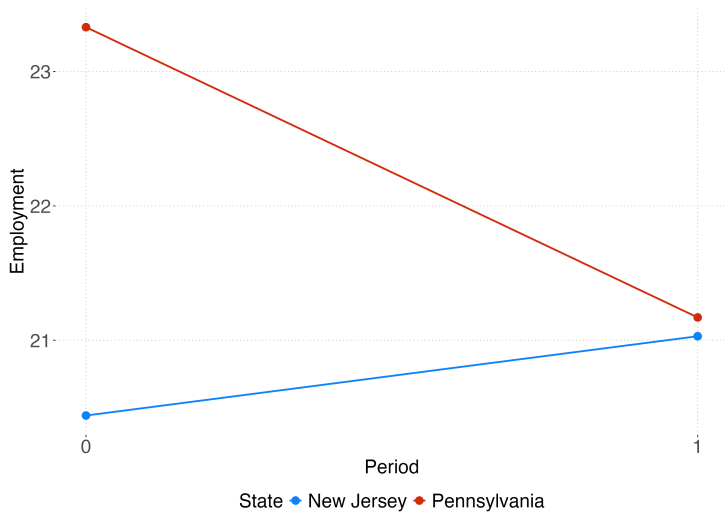
- Magnitude of the effects

$$|\tau^1| \geq |\tau^0| \implies \text{sgn}(DiD) = \text{sgn}(\tau^1).$$

Application

- Revisiting “Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania” Card and Krueger (1994).
- Effect of New Jersey’s minimum wage increase in 1992.
- Data on neighboring fast food restaurants in New Jersey and Pennsylvania.
- SUTVA unlikely to hold.
- the increase of minimum wages may also affect bordering regions in various ways
 - e.g., cross-border commuting: some Pennsylvania workers may have sought employment in New Jersey to benefit from higher wages
 - this may, in part, explain the observed decline in Pennsylvania employment

Application



What can we conclude?

- Plausible assumption here: $\tau^0 \leq 0$.
- This would imply that the *DiD* can be interpreted as an upper bound of τ^1 .
- Card and Krueger (1994) find an effect of 2.75 for New Jersey, but:
 - This is the upper bound, the effect on New Jersey could be small or even negative.
 - Sign of τ^1 identified if $|\tau^1| \geq |\tau^0|$

Concluding remarks and extensions for future research

- We have shown that, in the presence of unknown interference, the canonical DiD only identifies a contrast of causal effects
- We have provided assumptions under which researchers can (partially) identify policy-relevant causal effects in such settings.
- We have remained agnostic about the form interference allowing for arbitrary patterns of spillovers.
- An alternative approach is to directly model interference, as a recent and growing literature has begun to do
- We have considered only the case of two groups and two periods.
- Explore multiple periods and staggered adoption settings
 - Exploit alternative parallel trends assumptions to separately identify TATT and ASC
 - Consider interference across groups and over time (carryover effects)

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